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Module 3: Generation Planning and Reliability

PSM-104

M. Tech (EE)

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Comprehensive Class Notes

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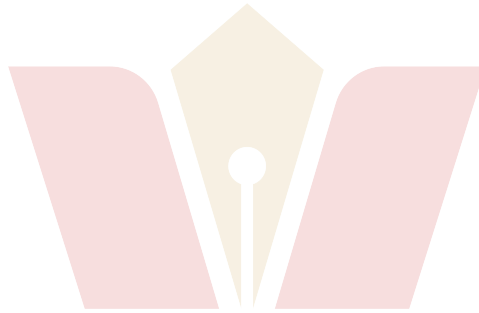
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1 Module 3 Overview: Generation Planning and Reliability

The primary aim of this module is to provide comprehensive understanding of generation system planning methodologies and reliability assessment techniques essential for ensuring adequate power supply in modern electrical systems. This module integrates theoretical foundations with practical applications, emphasizing the critical importance of reliable generation capacity in meeting energy demands while maintaining economic efficiency and environmental sustainability.

⚠ Module Philosophy: Generation planning represents the cornerstone of power system reliability, requiring sophisticated analysis of capacity adequacy, economic optimization, and risk management to ensure continuous electricity supply under diverse operating conditions and future uncertainties.

Modern generation systems face unprecedented challenges from renewable energy integration, market restructuring, environmental regulations, and evolving demand patterns. The methodologies covered in this module provide the analytical framework necessary to address these complexities while maintaining system adequacy and economic viability.

1.1 Module Learning Outcomes

Upon successful completion of this module, students will demonstrate the ability to:

1. **Evaluate** generation system reliability using advanced probabilistic methods including LOLP, LOLE, and EENS
2. **Analyze** capacity expansion strategies considering economic, technical, and environmental constraints
3. **Model** generation system behavior under scheduled and forced outage conditions
4. **Apply** integrated resource planning principles to optimize generation portfolios
5. **Assess** interconnected system benefits and reliability contributions
6. **Implement** computational algorithms for generation reliability evaluation
7. **Design** capacity expansion plans that balance reliability and economic objectives

1.2 Module Structure and Coverage

Table 1: Module 3 Detailed Topic Coverage

Lecture	Topic	Duration
3.1	Generation Planning Overview and Framework	50 min
3.2	Integrated Resource Planning Methodology	50 min
3.3	Generation System Modeling Techniques	50 min
3.4	Loss of Load Probability Analysis	50 min
3.5	Capacity Expansion Strategies	50 min
3.6	Scheduled and Forced Outage Analysis	50 min
3.7	Interconnected System Analysis	50 min

Table 1 – continued from previous page

Lecture	Topic	Duration
3.8	Emergency Assistance and Contingency Planning	50 min

2 Lecture 3.1: Generation Planning Overview and Framework

Generation planning determines the optimal mix, timing, and location of generating resources required to meet projected electricity demand while satisfying reliability criteria, economic objectives, and environmental constraints. This lecture establishes the foundational concepts and framework for systematic generation system planning.

2.1 Generation Planning Fundamentals

2.1.1 Planning Objectives and Scope

Generation planning pursues multiple, often competing objectives that must be systematically balanced:

- **Reliability Adequacy:** Ensuring sufficient generation capacity to meet demand with acceptable risk levels
- **Economic Efficiency:** Minimizing total system costs including capital, operating, and environmental expenses
- **Environmental Compliance:** Meeting emission standards and sustainability requirements
- **Fuel Diversity:** Reducing dependence on single fuel sources and associated risks
- **Flexibility:** Providing capability to accommodate load variability and renewable integration
- **System Security:** Maintaining adequate reserves and response capabilities

 **Generation Planning Definition:** The systematic process of determining the type, size, timing, and location of generating resources to be added to the power system over a specified planning horizon, while meeting reliability criteria, minimizing costs, and satisfying operational and environmental constraints.

2.1.2 Planning Time Horizons

Generation planning operates across multiple time scales, each requiring distinct methodologies and considerations:

Long-term Planning (5-20 years):

- Strategic capacity expansion decisions
- Technology selection and portfolio optimization
- Major infrastructure investments
- Regulatory compliance planning

- Load growth accommodation

Medium-term Planning (1-5 years):

- Detailed project development and financing
- Construction scheduling and resource allocation
- Regulatory approval processes
- Interconnection studies
- Fuel supply contract negotiations

Short-term Planning (months to 1 year):

- Maintenance scheduling optimization
- Seasonal resource adequacy assessment
- Fuel inventory management
- Capacity market participation
- Emergency preparedness planning

2.2 Generation Mix Optimization

2.2.1 Technology Characterization

Different generation technologies exhibit distinct characteristics that influence planning decisions:

Technology Type	Capital Cost (\$/kW)	O&M Cost (\$/MWh)	Flexibility (Start-up)	Emissions (kg CO ₂ /MWh)
Coal Steam	3,000-4,000	25-35	Low (hours)	900-1,000
CCGT	1,000-1,500	15-25	Medium (30-60 min)	350-450
OCGT	700-1,000	40-60	High (10-15 min)	550-650
Nuclear	5,000-7,000	10-15	Very Low (days)	10-20
Hydro	1,500-5,000	5-10	Very High (seconds)	0-10
Wind	1,300-1,800	10-15	Variable	0
Solar PV	1,000-1,500	5-10	Variable	0

2.2.2 Load Duration Curve Analysis

Load duration curves provide essential information for generation planning by representing the cumulative time that system load exceeds various levels throughout a specified period.

Load Duration Curve Construction:

1. Collect hourly load data for planning period (typically one year = 8,760 hours)
2. Sort load values in descending order from peak to minimum
3. Plot sorted load values versus cumulative time duration
4. Calculate energy area under curve for capacity factor analysis

The area under the load duration curve represents total energy demand:

$$\text{Total Energy} = \int_0^T L(t) dt \quad (1)$$

where $L(t)$ is the load level at duration t and T is the total period.

Generation Technology Matching:

Different portions of the load duration curve are optimally served by different generation technologies:

- **Baseload Region (70-100% duration):** Nuclear, coal, run-of-river hydro, geothermal
- **Intermediate Region (20-70% duration):** Combined cycle gas turbines, biomass
- **Peak Region (0-20% duration):** Open cycle gas turbines, pumped hydro storage, demand response

Example 2.1 (Load Duration Curve Analysis):

Consider a system with the following annual characteristics:

- Peak load: 5,000 MW
- Minimum load: 2,000 MW
- Annual energy consumption: 35,000 GWh
- Hours in year: 8,760

System Load Factor Calculation:

$$\text{Average Load} = \frac{35,000,000 \text{ MWh}}{8,760 \text{ hours}} = 3,995 \text{ MW} \quad (2)$$

$$\text{Load Factor} = \frac{\text{Average Load}}{\text{Peak Load}} = \frac{3,995}{5,000} = 0.799 = 79.9\% \quad (3)$$

Technology Allocation Strategy:

- Baseload capacity (2,000 MW at 95% CF): Nuclear and coal units
- Intermediate capacity (2,000 MW at 50% CF): CCGT units
- Peak capacity (1,000 MW at 10% CF): OCGT and storage

2.3 Economic Analysis Framework

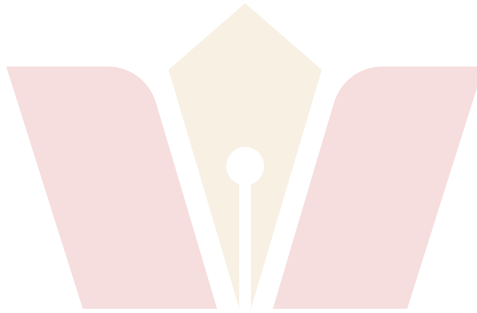
2.3.1 Levelized Cost of Energy

The levelized cost of energy (LCOE) provides a standardized metric for comparing generation technologies with different cost structures and operational characteristics:

$$\text{LCOE} = \frac{\sum_{t=0}^N \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=1}^N \frac{E_t}{(1+r)^t}} \quad (4)$$

where:

- I_t = Investment expenditure in year t
- M_t = Operations and maintenance expenditure in year t
- F_t = Fuel expenditure in year t
- E_t = Energy generation in year t
- r = Discount rate
- N = Economic lifetime of the project



\$ LCOE Calculation Example for CCGT Plant:

Given parameters:

- Capital cost: \$1,200/kW
- Plant capacity: 500 MW
- Capacity factor: 60%
- Fixed O&M: \$15/kW-year
- Variable O&M: \$3.5/MWh
- Heat rate: 6,800 Btu/kWh
- Natural gas price: \$4.00/MMBtu
- Economic life: 30 years
- Discount rate: 7%

Annual Calculations:

$$\text{Annual Energy} = 500 \times 0.60 \times 8,760 = 2,628,000 \text{ MWh} \quad (5)$$

$$\text{Capital Cost} = 500,000 \times 1,200 = \$600,000,000 \quad (6)$$

$$\text{Annual Fixed O\&M} = 500,000 \times 15 = \$7,500,000 \quad (7)$$

$$\text{Fuel Cost per MWh} = \frac{6,800 \times 4.00}{1,000} = \$27.20/\text{MWh} \quad (8)$$

$$\text{Annual Fuel Cost} = 2,628,000 \times 27.20 = \$71,481,600 \quad (9)$$

$$\text{Annual Variable O\&M} = 2,628,000 \times 3.5 = \$9,198,000 \quad (10)$$

LCOE Calculation: Using capital recovery factor and present value calculations with 7% discount rate over 30 years:

$$\text{Capital Recovery Factor} = \frac{r(1+r)^N}{(1+r)^N - 1} = \frac{0.07(1.07)^{30}}{(1.07)^{30} - 1} = 0.0806 \quad (11)$$

$$\text{Annualized Capital} = 600,000,000 \times 0.0806 = \$48.36 \text{ million/year} \quad (12)$$

$$\text{Annual Fixed O\&M} = \$7.5 \text{ million/year} \quad (13)$$

$$\text{Annual Fuel Cost} = \$71.48 \text{ million/year} \quad (14)$$

$$\text{Annual Variable O\&M} = \$9.20 \text{ million/year} \quad (15)$$

$$\text{Total Annual Cost} = 48.36 + 7.5 + 71.48 + 9.20 = \$136.54 \text{ million/year} \quad (16)$$

$$\text{Annual Energy} = 2,628,000 \text{ MWh} \quad (17)$$

$$\text{LCOE} = \frac{136,540,000}{2,628,000} = \$51.95/\text{MWh} \quad (18)$$

2.3.2 Screening Curve Analysis

Screening curves graphically represent the total annual cost of different generation technologies as a function of capacity factor, facilitating technology selection decisions.

Total Annual Cost Formulation:

$$\text{TAC}(CF) = \text{Annualized Capital Cost} + \text{Fixed O\&M} + CF \times 8,760 \times \text{Variable Cost} \quad (19)$$

where:

- TAC = Total Annual Cost per kW of capacity
- CF = Capacity Factor (fraction of time operating)
- Variable Cost = Fuel cost + Variable O&M (\$/MWh)

Technology Selection Principles:

- Technologies with lower capital costs but higher variable costs are economical for low capacity factors (peaking duty)
- Technologies with higher capital costs but lower variable costs are economical for high capacity factors (baseload duty)
- The intersection points of screening curves define capacity factor thresholds for technology transitions

3 Lecture 3.2: Integrated Resource Planning Methodology

Integrated Resource Planning (IRP) represents a comprehensive planning approach that evaluates both supply-side (generation) and demand-side (energy efficiency, demand response) resources on a consistent basis to develop optimal resource portfolios that meet system needs while considering multiple objectives and uncertainties.

3.1 IRP Framework and Philosophy

3.1.1 Core Principles of IRP

The IRP framework rests on several fundamental principles that distinguish it from traditional supply-side planning:

1. **Resource Neutrality:** Supply and demand-side resources evaluated using consistent criteria
2. **Multi-objective Optimization:** Consideration of cost, reliability, environmental, and social objectives
3. **Stakeholder Engagement:** Incorporation of diverse perspectives through public participation
4. **Uncertainty Analysis:** Explicit treatment of uncertainties through scenario and sensitivity analysis
5. **Portfolio Optimization:** Development of diverse resource portfolios that perform well across scenarios
6. **Action Plan Development:** Translation of long-term plans into near-term implementation actions

 **Integrated Resource Planning:** A systematic decision-making process that evaluates the full range of supply and demand-side resource options to meet current and future customer energy service needs at the lowest total cost while satisfying reliability, environmental, and other societal objectives.

3.1.2 IRP Process Steps

The IRP process typically follows a structured sequence of analytical steps:

1. Load Forecasting and Characterization

- Develop base, high, and low load growth scenarios
- Characterize load profiles and variability patterns
- Identify load flexibility and demand response potential
- Assess impacts of energy efficiency programs

2. Resource Assessment

- Catalog available supply-side resources and characteristics
- Evaluate demand-side management program opportunities
- Assess renewable energy resource potential and variability
- Identify transmission and distribution alternatives

3. Resource Screening and Portfolio Development

- Screen resources based on preliminary economic and technical criteria
- Develop candidate resource portfolios with different characteristics
- Apply optimization methods to identify efficient portfolios
- Consider resource diversity and portfolio risk characteristics

4. Portfolio Evaluation and Comparison

- Evaluate portfolios across multiple scenarios and metrics
- Conduct cost-effectiveness analysis using present value methods
- Assess reliability performance using probabilistic methods
- Evaluate environmental impacts and compliance

5. Risk and Uncertainty Analysis

- Perform sensitivity analysis on key uncertainties
- Conduct scenario analysis for major decision drivers
- Apply risk metrics to portfolio comparisons
- Identify robust strategies across uncertain futures

6. Action Plan and Implementation

- Develop near-term action plan from long-term strategy
- Establish monitoring and evaluation framework
- Define decision points and flexibility options
- Create stakeholder communication and engagement plan

3.2 Demand-Side Resource Integration

3.2.1 Energy Efficiency Programs

Energy efficiency represents the most cost-effective resource in many planning studies, reducing both energy consumption and peak demand.

Energy Efficiency Resource Characterization:

$$\text{Avoided Supply Cost} = \Delta E \times \text{LCOE}_{\text{marginal}} + \Delta P_{\text{peak}} \times \text{Capacity Cost} \quad (20)$$

where:

- ΔE = Annual energy savings (MWh)
- ΔP_{peak} = Peak demand reduction (MW)
- $\text{LCOE}_{\text{marginal}}$ = Marginal cost of generation
- Capacity Cost = Cost of capacity resources (\$/kW-year)

Cost-Effectiveness Tests:

Multiple perspectives exist for evaluating energy efficiency program cost-effectiveness:

Test	Costs Included	Benefits Included
Participant Cost Test	Customer investment and bill impacts	Customer energy savings and incentives
Ratepayer Impact Test	Utility program costs and lost revenues	Avoided supply costs
Total Resource Cost Test	Utility and customer costs	Avoided supply costs
Societal Cost Test	All costs including externalities	All benefits including externalities

Example 3.1 (Energy Efficiency Program Evaluation):

Consider a residential lighting efficiency program with the following characteristics:

Program Parameters:

- Number of participants: 50,000 households
- LED bulbs per household: 15
- Energy savings per bulb: 60 kWh/year
- Bulb lifetime: 15 years
- Total program cost: \$2,000,000
- Customer contribution: 30%
- Peak demand reduction per bulb: 0.055 kW
- Peak coincidence factor: 0.15

Program Impact Calculations:

$$\text{Annual Energy Savings} = 50,000 \times 15 \times 60 = 45,000,000 \text{ kWh} = 45 \text{ GWh} \quad (21)$$

$$\text{Lifetime Energy Savings} = 45 \times 15 = 675 \text{ GWh} \quad (22)$$

$$\text{Peak Demand Reduction} = 50,000 \times 15 \times 0.055 \times 0.15 = 6.19 \text{ MW} \quad (23)$$

Cost-Effectiveness Assessment: Assuming avoided energy cost of \$60/MWh and capacity cost of \$100/kW-year:

$$\text{Lifetime Avoided Energy Cost} = 675,000 \times 60 = \$40,500,000 \quad (24)$$

$$\text{Annual Avoided Capacity Cost} = 6,190 \times 100 = \$619,000 \quad (25)$$

$$\text{Lifetime Avoided Capacity Cost} = 619,000 \times 15 = \$9,285,000 \quad (26)$$

$$\text{Total Avoided Costs} = 40,500,000 + 9,285,000 = \$49,785,000 \quad (27)$$

$$\text{Benefit-Cost Ratio} = \frac{49,785,000}{2,000,000} = 24.9 \quad (28)$$

The program demonstrates exceptional cost-effectiveness with a benefit-cost ratio exceeding 24:1.

3.2.2 Demand Response Resources

Demand response programs provide system operators with the ability to modify customer electricity consumption patterns in response to system conditions or price signals.

Demand Response Categories:

- **Emergency/Reliability DR:** Load reductions during system emergencies (few hours per year)
- **Economic/Price-responsive DR:** Load modifications in response to wholesale price signals
- **Ancillary Services DR:** Provision of reserves, regulation, and other grid services
- **Time-of-Use Programs:** Load shifting encouraged through time-differentiated rates

Demand Response Valuation:

$$\text{DR Value} = \sum_{i=1}^N P_i \times (\text{Capacity Cost}_i + \Delta \text{Energy Cost}_i \times H_i) - \text{Program Cost} \quad (29)$$

where:

- P_i = DR capacity available in scenario i (MW)
- Capacity Cost_i = Avoided generation capacity cost (\$/kW)
- $\Delta \text{Energy Cost}_i$ = Energy cost differential during events (\$/MWh)
- H_i = Event hours in scenario i
- Program Cost = Total DR program implementation and incentive costs

3.3 Multi-Criteria Decision Analysis

3.3.1 Decision Criteria and Metrics

IRP evaluates resource portfolios using multiple performance metrics spanning economic, reliability, environmental, and risk dimensions:

Economic Metrics:

- Net Present Value (NPV) of total costs

- Levelized Cost of Energy (LCOE)
- Customer rate impacts over time
- Economic development benefits

Reliability Metrics:

- Loss of Load Probability (LOLP)
- Expected Unserved Energy (EUE)
- Planning reserve margin adequacy
- System flexibility indices

Environmental Metrics:

- CO₂ emissions (tons/year)
- Criteria air pollutants (NO_x, SO₂, PM)
- Water consumption intensity
- Land use impacts

Risk and Uncertainty Metrics:

- Cost volatility across scenarios
- Fuel price exposure and hedging
- Regulatory compliance risk
- Technology performance uncertainty

💡 Portfolio Scoring and Ranking:

Many IRP processes employ multi-attribute utility theory or weighted scoring methods to aggregate diverse metrics into composite portfolio scores:

$$\text{Portfolio Score} = \sum_{j=1}^M w_j \times U_j(x_j) \quad (30)$$

where:

- w_j = Weight for criterion j (reflecting stakeholder preferences)
- $U_j(x_j)$ = Utility function mapping criterion value to normalized score
- M = Number of evaluation criteria

Weights typically sum to unity: $\sum_{j=1}^M w_j = 1$

4 Lecture 3.3: Generation System Modeling Techniques

Accurate modeling of generation system behavior under various operating conditions forms the foundation of reliability assessment and planning studies. This lecture covers probabilistic modeling techniques essential for capturing the stochastic nature of generation equipment failures and repairs.

4.1 Component Reliability Models

4.1.1 Two-State Model

The fundamental representation of generating unit reliability employs a two-state model capturing operational (up) and failed (down) states.

⚙️ Two-State Generation Unit Model:

State definitions:

- **State 1 (Up):** Unit available for generation at rated capacity
- **State 0 (Down):** Unit unavailable due to forced outage or maintenance

State probabilities:

$$P(\text{Up}) = A = \frac{\mu}{\lambda + \mu} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} \quad (31)$$

$$P(\text{Down}) = U = \frac{\lambda}{\lambda + \mu} = \frac{\text{MTTR}}{\text{MTTF} + \text{MTTR}} \quad (32)$$

where:

- λ = Failure rate (failures/hour)
- μ = Repair rate (repairs/hour)
- A = Availability
- U = Unavailability (Forced Outage Rate)

Typical Generating Unit FOR Values:

Unit Type	Typical FOR	MTTF (hours)	MTTR (hours)
Coal Steam	5-8%	1,200-1,500	60-100
CCGT	3-5%	2,000-2,500	50-80
OCGT	2-4%	3,000-4,000	40-60
Nuclear	2-5%	1,500-2,000	50-150
Hydro	1-3%	4,000-6,000	30-50
Wind Turbine	3-7%	1,500-2,000	40-100
Solar PV	1-2%	8,000-10,000	20-40

Example 4.1 (Two-State Model Calculation):

Consider a 500 MW coal-fired generating unit with the following reliability characteristics:

- Mean Time To Failure (MTTF) = 1,400 hours
- Mean Time To Repair (MTTR) = 75 hours

Calculate availability and forced outage rate:

$$\text{Availability (A)} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} = \frac{1,400}{1,400 + 75} = \frac{1,400}{1,475} = 0.9492 = 94.92\% \quad (33)$$

$$\text{FOR (U)} = \frac{\text{MTTR}}{\text{MTTF} + \text{MTTR}} = \frac{75}{1,475} = 0.0508 = 5.08\% \quad (34)$$

$$\text{Expected Available Capacity} = 500 \times 0.9492 = 474.6 \text{ MW} \quad (35)$$

$$\text{Expected Unavailable Capacity} = 500 \times 0.0508 = 25.4 \text{ MW} \quad (36)$$

Alternative calculation using failure and repair rates:

$$\lambda = \frac{1}{\text{MTTF}} = \frac{1}{1,400} = 0.000714 \text{ failures/hour} \quad (37)$$

$$\mu = \frac{1}{\text{MTTR}} = \frac{1}{75} = 0.01333 \text{ repairs/hour} \quad (38)$$

$$A = \frac{\mu}{\lambda + \mu} = \frac{0.01333}{0.000714 + 0.01333} = 0.9492 \quad (39)$$

4.1.2 Multi-State Capacity Outage Model

Many generating units can operate at reduced capacity levels due to partial outages of subsystems or equipment derating conditions. Multi-state models capture this operational reality more accurately.

Three-State Model Example:

- **State 2:** Full capacity operation (probability p_2)
- **State 1:** Partial capacity operation at $\alpha \times$ capacity (probability p_1)
- **State 0:** Complete outage (probability p_0)

where $p_0 + p_1 + p_2 = 1$ and $0 < \alpha < 1$.

Expected Available Capacity:

$$E[C] = C_{\text{rated}} \times (p_2 + \alpha \times p_1) \quad (40)$$

4.2 System Capacity Model Development

4.2.1 Capacity Outage Probability Table (COPT)

The COPT represents the probability distribution of total system capacity in different outage states, formed by combining individual unit outage probabilities.

COPT Construction Algorithm:

For a system with N generating units:

1. Initialize COPT with State 0 (no capacity) at probability 1.0
2. For each unit i from 1 to N :
 - Create temporary table with existing states
 - For each existing state with capacity C_j and probability P_j :
 - Add state $(C_j, P_j \times U_i)$ if unit i fails
 - Add state $(C_j + C_i, P_j \times A_i)$ if unit i operates
 - Combine states with identical capacity, summing probabilities
 - Update COPT with new state table
3. Sort final COPT by capacity level
4. Calculate cumulative probabilities

Example 4.2 (COPT Construction for Three-Unit System):

Consider a generation system with three identical units:

- Unit capacity: 100 MW each
- Availability: 0.95
- FOR: 0.05

Systematic COPT Construction:

Initial state (no units):

Capacity (MW)	Probability
0	1.000

After adding Unit 1:

Capacity (MW)	Probability	Calculation
0	0.050	1.0×0.05
100	0.950	1.0×0.95

After adding Unit 2:

Capacity (MW)	Probability	Calculation
0	0.0025	0.05×0.05
100	0.0975	$0.05 \times 0.95 + 0.95 \times 0.05$
200	0.9025	0.95×0.95

After adding Unit 3 (Final COPT):

Capacity (MW)	Individual Prob	Cumulative Prob	States
0	0.000125	0.000125	All units out
100	0.007125	0.007250	Two units out
200	0.135375	0.142625	One unit out
300	0.857375	1.000000	All units operating

Verification:

$$\text{Sum of probabilities} = 0.000125 + 0.007125 + 0.135375 + 0.857375 = 1.000\checkmark \quad (41)$$

$$\text{Expected capacity} = 0 \times 0.000125 + 100 \times 0.007125 \quad (42)$$

$$+ 200 \times 0.135375 + 300 \times 0.857375 \quad (43)$$

$$= 0 + 0.7125 + 27.075 + 257.2125 = 285 \text{ MW} \quad (44)$$

$$\text{Alternative calculation} = 3 \times 100 \times 0.95 = 285 \text{ MW}\checkmark \quad (45)$$

4.2.2 Recursive Algorithm for Large Systems

For systems with many generating units, the recursive algorithm provides computational efficiency by avoiding explicit enumeration of all possible states.

Recursive Formulation:

$$P_n(C) = P_{n-1}(C) \times U_n + P_{n-1}(C - C_n) \times A_n \quad (46)$$

where:

- $P_n(C)$ = Probability of exactly capacity C with n units considered
- $P_{n-1}(C)$ = Probability of capacity C with $(n - 1)$ units
- C_n = Capacity of unit n
- A_n = Availability of unit n
- U_n = Unavailability (FOR) of unit n

Algorithm Implementation Steps:

1. Initialize: $P_0(0) = 1.0$, all other $P_0(C) = 0$
2. For unit $n = 1$ to N :
 - For capacity $C = C_{\text{total}}$ down to C_n :
 - $P_n(C) = P_{n-1}(C) \times U_n + P_{n-1}(C - C_n) \times A_n$
 - For capacity $C = C_n - 1$ down to 0:
 - $P_n(C) = P_{n-1}(C) \times U_n$
3. Output final probability distribution $P_N(C)$

4.3 Load Modeling for Reliability Studies**4.3.1 Deterministic Load Model**

The simplest approach assumes load takes a single fixed value (typically peak load) with certainty.

Peak Load Model:

$$L(t) = L_{\text{peak}} \quad \forall t \in \text{planning period} \quad (47)$$

This conservative approach overestimates reliability risk but provides computational simplicity.

4.3.2 Multi-Level Load Model

More realistic representation using load duration curve discretized into discrete levels:

Load Level Representation:

$$\text{Load Level } k : L_k \text{ MW for } T_k \text{ hours} \quad (48)$$

where $\sum_{k=1}^M T_k = 8,760$ hours (annual period).

Load Model Development from Historical Data:

1. Sort hourly loads in descending order
2. Divide into M bins of approximately equal duration
3. Assign average load in each bin as representative level
4. Record duration for each load level

Multi-Level Load Model Construction:

Annual load data summary:

- Peak load: 1,000 MW
- Average load: 700 MW
- Minimum load: 400 MW

Six-Level Load Model:

Level	Load (MW)	Hours/Year	Energy (GWh)
1	950	876	832.2
2	850	1,752	1,489.2
3	750	2,190	1,642.5
4	650	2,190	1,423.5
5	550	1,314	722.7
6	450	438	197.1
Total	-	8,760	6,307.2

Model Validation:

$$\text{Average Load from Model} = \frac{6,307,200}{8,760} = 720 \text{ MW} \quad (49)$$

$$\text{Error} = \frac{720 - 700}{700} = 2.86\% \text{ (acceptable)} \quad (50)$$

4.3.3 Probabilistic Load Model

Advanced studies incorporate load uncertainty through probability distributions:

Load Forecast Uncertainty Model:

$$L(t) = \hat{L}(t) \times (1 + \epsilon) \quad (51)$$

where:

- $\hat{L}(t)$ = Forecast load at time t
- ϵ = Random forecast error, typically $\epsilon \sim N(0, \sigma^2)$
- σ = Standard deviation of forecast error (2-5% for annual peak)

5 ❌ Lecture 3.4: Loss of Load Probability Analysis

Loss of Load Probability (LOLP) quantifies the probability that available generation capacity will be insufficient to meet system load demand, providing a fundamental reliability metric for generation system adequacy assessment.

5.1 LOLP Fundamentals

5.1.1 Basic Definitions and Concepts

! Loss of Load Probability (LOLP):

The probability that the available generation capacity is less than the system load demand at a given time:

$$\text{LOLP} = P(\text{Available Capacity} < \text{Load}) \quad (52)$$

Related reliability indices:

- **LOLE (Loss of Load Expectation):** Expected number of time periods (days or hours) when load exceeds capacity
- **LOLH (Loss of Load Hours):** Expected number of hours per year with insufficient capacity
- **EUE (Expected Unserved Energy):** Expected amount of energy not served due to capacity shortfalls (MWh)

Industry Reliability Standards:

Common planning reliability criteria used in practice:

Criterion	Target Value	Interpretation
LOLE	0.1 days/year	One day in ten years
LOLP (daily)	$0.1/365 = 0.000274$	Daily probability
LOLH	2.4 hours/year	Approximately 0.1 days
EUE	Varies by system	Typically < 0.01% of energy

5.2 LOLP Calculation Methods

5.2.1 Convolution Method

The convolution method systematically combines the COPT with load probability distribution to calculate LOLP.

Mathematical Formulation:

$$\text{LOLP} = \sum_{C=0}^{C_{\text{total}}} P_{\text{cap}}(C) \times P_{\text{load}}(\text{Load} > C) \quad (53)$$

where:

- $P_{\text{cap}}(C)$ = Probability of exactly capacity C available
- $P_{\text{load}}(\text{Load} > C)$ = Probability that load exceeds capacity C

For deterministic peak load L_{peak} :

$$\text{LOLP} = \sum_{C < L_{\text{peak}}} P_{\text{cap}}(C) \quad (54)$$

This represents the cumulative probability of capacity states insufficient to meet peak load.

Example 5.1 (LOLP Calculation with Deterministic Load):

Using the three-unit system from the previous example with peak load = 250 MW:

System COPT:

Available Capacity (MW)	Probability	Adequate?
0	0.000125	No
100	0.007125	No
200	0.135375	No
300	0.857375	Yes

LOLP Calculation:

$$\text{LOLP} = P(C < 250) \quad (55)$$

$$= P(C = 0) + P(C = 100) + P(C = 200) \quad (56)$$

$$= 0.000125 + 0.007125 + 0.135375 \quad (57)$$

$$= 0.142625 \quad (58)$$

Interpretation:

- $\text{LOLP} = 0.142625 = 14.26\%$
- $\text{LOLE} = 0.142625 \times 365 = 52.06 \text{ days/year}$
- This system does NOT meet the 0.1 days/year criterion

Required Capacity Addition: To meet reliability criterion, need sufficient capacity such that:

$$P(C < L_{\text{peak}}) \leq \frac{0.1}{365} = 0.000274 \quad (59)$$

This would require adding another unit or reducing peak load through demand-side measures.

5.2.2 Multi-Level Load Model LOLP

When using a multi-level load representation, LOLP is calculated for each load level and weighted by duration:

Annual LOLP Formulation:

$$\text{LOLP}_{\text{annual}} = \frac{1}{8,760} \sum_{k=1}^M T_k \times \text{LOLP}_k \quad (60)$$

where:

- M = Number of load levels
- T_k = Duration of load level k (hours)
- LOLP_k = Probability of capacity inadequacy at load level k

Load Level LOLP:

$$\text{LOLP}_k = \sum_{C < L_k} P_{\text{cap}}(C) \quad (61)$$

5.3 Expected Unserved Energy (EUE)

5.3.1 EUE Calculation Framework

EUE quantifies the expected magnitude of capacity shortfalls, providing additional insight beyond probability-based metrics.

General EUE Formulation:

$$\text{EUE} = \sum_{C=0}^{C_{\text{total}}} P_{\text{cap}}(C) \times E[(L - C)^+] \quad (62)$$

where $(L - C)^+ = \max(0, L - C)$ represents the capacity shortfall when load exceeds capacity.

For deterministic load:

$$\text{EUE} = \sum_{C < L} P_{\text{cap}}(C) \times (L - C) \quad (63)$$

For multi-level load model:

$$\text{EUE}_{\text{annual}} = \sum_{k=1}^M T_k \sum_{C < L_k} P_{\text{cap}}(C) \times (L_k - C) \quad (64)$$

Example 5.2 (EUE Calculation):

Using the three-unit system (peak load = 250 MW):

Capacity shortfall for each state:

Capacity (MW)	Probability	Shortfall (MW)	Weighted Shortfall
0	0.000125	250	0.031
100	0.007125	150	1.069
200	0.135375	50	6.769
300	0.857375	0	0.000
Total EUE (MW):			7.869

If this represents daily assessment:

$$\text{EUE}_{\text{daily}} = 7.869 \text{ MW-hours/day} \quad (65)$$

$$\text{EUE}_{\text{annual}} = 7.869 \times 365 = 2,872 \text{ MW-hours/year} = 2.87 \text{ GWh/year} \quad (66)$$

As percentage of total energy: Assuming 250 MW average load:

$$\text{Total Energy} = 250 \times 8,760 = 2,190,000 \text{ MWh} \quad (67)$$

$$\text{EUE Percentage} = \frac{2,872}{2,190,000} = 0.131\% \text{ of annual energy} \quad (68)$$

5.3.2 Frequency and Duration Approach

Advanced reliability assessment considers not only the probability of capacity shortfalls but also their expected frequency and duration.

Loss of Load Frequency:

$$f_{\text{LOL}} = \sum_{C < L} \lambda_{\text{system}}(C) \times P_{\text{cap}}(C) \quad (69)$$

where $\lambda_{\text{system}}(C)$ is the transition rate from adequate to inadequate states.

Average Duration of Loss of Load:

$$d_{\text{LOL}} = \frac{\text{LOLE}}{f_{\text{LOL}}} \quad (70)$$

Relationship to LOLP:

$$\text{LOLP} = f_{\text{LOL}} \times d_{\text{LOL}} \quad (71)$$

This frequency-duration decomposition provides valuable insights for planning decisions and operating procedures.

6 [] Lecture 3.5: Capacity Expansion Strategies

Capacity expansion planning determines the optimal timing, type, and magnitude of generation additions to meet growing demand while maintaining reliability and minimizing costs. This lecture covers strategic frameworks and optimization approaches for capacity expansion decisions.

6.1 Capacity Expansion Problem Formulation

6.1.1 Dynamic Optimization Framework

The capacity expansion problem can be formulated as a multi-period optimization problem:

Objective Function:

$$\min \sum_{t=1}^T \frac{1}{(1+r)^t} \left[\sum_i IC_{i,t} x_{i,t} + \sum_i FC_{i,t} y_{i,t} + \sum_i VC_{i,t} E_{i,t} \right] \quad (72)$$

where:

- T = Planning horizon (years)
- r = Discount rate
- $IC_{i,t}$ = Investment cost for technology i in year t (\$/kW)
- $FC_{i,t}$ = Fixed operating cost for technology i (\$/kW-year)
- $VC_{i,t}$ = Variable operating cost (\$/MWh)
- $x_{i,t}$ = Capacity addition of technology i in year t (MW)
- $y_{i,t}$ = Total installed capacity of technology i in year t (MW)
- $E_{i,t}$ = Energy generation from technology i in year t (MWh)

Key Constraints:

1. Capacity Evolution:

$$y_{i,t} = y_{i,t-1} + x_{i,t} - r_{i,t} \quad (73)$$

where $r_{i,t}$ represents retirements.

2. Energy Balance:

$$\sum_i E_{i,t} \geq D_t \quad (74)$$

where D_t is demand in year t .

3. Reliability Constraint:

$$\sum_i y_{i,t} \times CC_i \geq L_{\text{peak},t} \times (1 + RM_{\text{target}}) \quad (75)$$

where:

- CC_i = Capacity credit of technology i
- RM_{target} = Target planning reserve margin

4. Generation Limits:

$$E_{i,t} \leq y_{i,t} \times CF_i \times 8,760 \quad (76)$$

where CF_i is the capacity factor.

5. Technology Penetration Limits:

$$y_{i,t} \leq \alpha_i \times \sum_j y_{j,t} \quad (77)$$

where α_i represents maximum share for technology i (often applied to variable renewables).

6.2 Planning Reserve Margin**6.2.1 Reserve Margin Concepts****🔒 Planning Reserve Margin:**

The installed capacity in excess of peak demand, expressed as percentage:

$$\text{Reserve Margin} = \frac{\text{Installed Capacity} - \text{Peak Demand}}{\text{Peak Demand}} \times 100\% \quad (78)$$

Typical values range from 10-20

- Generation fleet reliability characteristics
- Load forecast uncertainty
- Interconnection capacity and assistance
- Renewable energy penetration
- Demand response capability

Capacity Credit Adjustments:

Not all capacity contributes equally to reliability. Capacity credits adjust for technology-specific characteristics:

Technology	Capacity Credit	Considerations
Baseload (Nuclear, Coal)	90-95%	High FOR reduces contribution
CCGT	90-95%	Reliable with good availability
OCGT	95-98%	High reliability, low FOR
Wind	10-40%	Depends on correlation with peak
Solar PV	10-70%	High if peak is mid-day
Hydro	80-95%	Depends on water availability
Storage	90-100%	Limited by duration

Effective Load Carrying Capability (ELCC):

More rigorous capacity credit assessment uses ELCC methodology:

$$\text{ELCC} = \Delta L \text{ such that } \text{LOLP}_{\text{before}} = \text{LOLP}_{\text{after addition}} \quad (79)$$

where the reliability level is maintained constant before and after resource addition.

6.3 Expansion Timing Optimization**6.3.1 Capacity Addition Staging**

Optimal timing balances early investment costs against reliability risks and future uncertainty.

Present Value Cost Comparison:

Compare present value of costs for different timing strategies:

$$\text{PV}_{\text{early}} = \frac{IC}{(1+r)^{t_1}} + \sum_{t=t_1}^T \frac{FC + VC \times CF \times 8,760}{(1+r)^t} \quad (80)$$

$$\text{PV}_{\text{late}} = \frac{IC}{(1+r)^{t_2}} + \sum_{t=t_2}^T \frac{FC + VC \times CF \times 8,760}{(1+r)^t} + \sum_{t=t_1}^{t_2-1} \frac{C_{\text{penalty}}}{(1+r)^t} \quad (81)$$

where C_{penalty} represents reliability penalty costs or emergency purchases.

Example 6.1 (Capacity Addition Timing Analysis):

Compare adding 500 MW CCGT unit in 2026 vs. 2028:

Economic Parameters:

- Capital cost: \$1,200/kW = \$600 million
- Fixed O&M: \$15/kW-year = \$7.5 million/year
- Variable O&M: \$3.5/MWh
- Capacity factor: 60%
- Economic life: 30 years
- Discount rate: 7%
- Expected reliability penalty 2026-2027: \$20 million/year if delayed

Present Value Analysis (Base Year 2025):**Option 1: Add in 2026**

$$\text{PV}_{\text{capital}} = \frac{600}{(1.07)^1} = \$560.7 \text{ million} \quad (82)$$

$$\text{PV}_{\text{fixed O\&M}} = 7.5 \times \sum_{t=1}^{30} \frac{1}{(1.07)^t} = 7.5 \times 12.41 = \$93.1 \text{ million} \quad (83)$$

$$\text{PV}_{\text{total}} = 560.7 + 93.1 + \text{PV variable costs} = \$653.8 \text{ million} \quad (84)$$

Option 2: Add in 2028

$$\text{PV}_{\text{capital}} = \frac{600}{(1.07)^3} = \$489.7 \text{ million} \quad (85)$$

$$\text{PV}_{\text{penalty costs}} = \frac{20}{(1.07)^1} + \frac{20}{(1.07)^2} = 18.7 + 17.5 = \$36.2 \text{ million} \quad (86)$$

$$\text{PV}_{\text{fixed O\&M}} = 7.5 \times \sum_{t=3}^{30} \frac{1}{(1.07)^t} = 7.5 \times 10.71 = \$80.3 \text{ million} \quad (87)$$

$$\text{PV}_{\text{total}} = 489.7 + 36.2 + 80.3 = \$606.2 \text{ million} \quad (88)$$

Conclusion: Delaying until 2028 saves approximately \$47.6 million in present value terms, provided reliability penalties are manageable through alternative measures (demand response, power purchases).

6.3.2 Lead Time Considerations

Generation project development involves significant lead times that constrain expansion timing:

Technology	Planning-to-Operation	Construction Period
OCGT	2-3 years	12-18 months
CCGT	3-4 years	24-30 months
Coal Steam	4-6 years	36-48 months
Nuclear	8-12 years	60-84 months
Wind Farm	2-4 years	12-24 months
Solar Farm	2-3 years	12-18 months
Pumped Hydro	5-8 years	48-72 months

Decision Point Analysis:

Given lead times, capacity expansion decisions must be made years before actual need:

$$t_{\text{decision}} = t_{\text{need}} - \text{Lead Time} \quad (89)$$

This introduces forecast uncertainty that must be managed through flexible strategies and option value analysis.

6.4 Technology Mix Optimization

6.4.1 Portfolio Optimization Approach

Modern capacity expansion increasingly uses portfolio optimization methods that balance expected costs against risk:

Mean-Variance Optimization:

$$\min \quad \mu_{\text{cost}} + \lambda \sigma_{\text{cost}}^2 \quad (90)$$

$$\text{s.t.} \quad \text{Reliability constraints} \quad (91)$$

$$\text{Capacity evolution constraints} \quad (92)$$

$$\text{Technology limits} \quad (93)$$

where:

- μ_{cost} = Expected cost across scenarios
- σ_{cost}^2 = Cost variance across scenarios
- λ = Risk aversion parameter

Efficient Frontier Construction:

Generate set of portfolios that minimize cost for different risk levels:

1. Define scenarios for key uncertainties (fuel prices, demand, technology costs)
2. Calculate portfolio cost under each scenario
3. Compute mean and variance of costs
4. Optimize for different values of λ to trace efficient frontier
5. Present decision makers with cost-risk tradeoffs

6.4.2 Fuel Diversity Objectives

Capacity expansion planning often includes explicit fuel diversity objectives to reduce exposure to fuel price volatility and supply disruptions.

Fuel Diversity Metrics:

Herfindahl-Hirschman Index (HHI):

$$\text{HHI} = \sum_{f=1}^F \left(\frac{E_f}{E_{\text{total}}} \right)^2 \times 10,000 \quad (94)$$

where E_f is energy from fuel type f .

- $\text{HHI} < 1,500$: Competitive/diverse fuel mix
- $1,500 < \text{HHI} < 2,500$: Moderately concentrated
- $\text{HHI} > 2,500$: Highly concentrated

Diversity Constraint:

$$\frac{E_f}{E_{\text{total}}} \leq \beta_f \quad \forall \text{ fuels } f \quad (95)$$

where β_f represents maximum allowable share (often 50-60%).

7 ✂ Lecture 3.6: Scheduled and Forced Outage Analysis

Generation system reliability assessment must consider both forced (random) outages and scheduled (planned) maintenance outages. This lecture covers methodologies for incorporating maintenance scheduling into reliability analysis and optimizing maintenance plans.

7.1 Outage Classification and Characteristics

7.1.1 Outage Type Definitions

Generation Unit Outage Types:

1. Forced Outage: Immediate removal from service due to equipment failure or operating limit violation. Characteristics:

- Random occurrence (Poisson or exponential process)
- Unpredictable timing
- Variable duration depending on failure severity
- Quantified by Forced Outage Rate (FOR)

2. Scheduled (Planned) Outage: Predetermined removal from service for maintenance, inspection, or overhaul. Characteristics:

- Predictable timing (scheduled in advance)
- Known duration
- Can be optimally scheduled
- Typically annual or periodic

3. Partial Outage: Operation at reduced capacity due to equipment degradation or environmental limits. Characteristics:

- May be forced or scheduled
- Requires multi-state modeling
- Common in large thermal units

Typical Maintenance Requirements:

Unit Type	Frequency	Duration	Annual Unavailability
Coal Steam	Annual	4-6 weeks	8-12%
CCGT	3-4 years	2-4 weeks	2-4%
OCGT	2-3 years	1-2 weeks	1-2%
Nuclear	18-24 months	3-6 weeks	6-10%
Hydro	Annual	1-2 weeks	2-4%
Wind	Semi-annual	1-3 days	<1%

7.2 Maintenance Scheduling Optimization

7.2.1 Maintenance Scheduling Problem Formulation

The maintenance scheduling problem determines optimal timing for planned outages while maintaining adequate system reliability and minimizing costs.

Objective Function:

$$\min \sum_{t=1}^T \sum_{i=1}^N C_{\text{replacement}}(i, t) + \sum_{t=1}^T C_{\text{production}}(t) + w \times \sum_{t=1}^T R_{\text{penalty}}(t) \quad (96)$$

where:

- $C_{\text{replacement}}(i, t)$ = Cost of replacement power for unit i maintenance in period t
- $C_{\text{production}}(t)$ = System production cost in period t
- $R_{\text{penalty}}(t)$ = Reliability penalty in period t
- w = Weight factor for reliability

Decision Variables:

$$x_{i,t} = \begin{cases} 1 & \text{if unit } i \text{ starts maintenance in period } t \\ 0 & \text{otherwise} \end{cases} \quad (97)$$

Key Constraints:

1. Maintenance Requirement:

$$\sum_{t=1}^T x_{i,t} = 1 \quad \forall i \quad (98)$$

Each unit must be maintained exactly once during planning horizon.

2. Maintenance Duration:

$$\sum_{s=t}^{t+d_i-1} u_{i,s} = d_i \times x_{i,t} \quad \forall i, t \quad (99)$$

where $u_{i,s}$ indicates unit i is under maintenance in period s , and d_i is maintenance duration.

3. Reliability Constraint:

$$\sum_{i=1}^N C_i(1 - u_{i,t}) \geq L_t(1 + RM_{\min}) \quad \forall t \quad (100)$$

Available capacity must meet load plus minimum reserve margin.

4. Crew/Resource Constraints:

$$\sum_{i=1}^N u_{i,t} \times R_{i,\text{crew}} \leq R_{\text{crew}}^{\max} \quad \forall t \quad (101)$$

Limited maintenance crews or specialized resources.

7.2.2 Heuristic Scheduling Rules

Practical maintenance scheduling often employs heuristic rules based on system characteristics:

Load-Based Scheduling:

- Schedule maintenance during low-load periods (typically spring and fall)
- Avoid maintenance during peak load seasons (summer and winter)
- Stagger maintenance to maintain adequate available capacity

Priority-Based Scheduling:

1. Calculate priority index for each unit: $PI_i = \frac{\text{Reserve}_{\text{ave}} - C_i}{\text{Maintenance Duration}_i}$
2. Schedule units with lowest priority index first

3. Check reliability constraint before confirming schedule
4. Adjust schedule if constraints violated

Example 7.1 (Maintenance Scheduling Problem):

Consider a six-unit system with annual maintenance requirements:

System Data:

Unit	Capacity (MW)	FOR	Maint. Duration (weeks)
1	400	0.05	5
2	400	0.05	5
3	300	0.04	4
4	300	0.04	4
5	200	0.03	3
6	200	0.03	3
Total	1,800	-	-

Load Profile (Seasonal Peaks):

- Winter (Weeks 1-13): 1,400 MW peak
- Spring (Weeks 14-26): 1,000 MW peak
- Summer (Weeks 27-39): 1,500 MW peak
- Fall (Weeks 40-52): 1,100 MW peak

Maintenance Schedule Optimization:

Constraint: Available capacity $\geq$ Peak load + 15% reserve margin

Optimal Schedule:

Unit	Maintenance Period	Season	Available Capacity
1	Weeks 15-19	Spring	1,400 MW
2	Weeks 42-46	Fall	1,400 MW
3	Weeks 20-23	Spring	1,500 MW
4	Weeks 47-50	Fall	1,500 MW
5	Weeks 24-26	Spring	1,600 MW
6	Weeks 40-42	Fall	1,600 MW

Verification:

$$\text{Minimum Available (Spring)} = 1,800 - 400 = 1,400 \text{ MW} \quad (102)$$

$$\text{Required Capacity (Spring)} = 1,000 \times 1.15 = 1,150 \text{ MW} \quad (103)$$

$$\text{Margin} = 1,400 - 1,150 = 250 \text{ MW} \checkmark \quad (104)$$

All seasonal constraints satisfied with adequate reserve margin.

7.3 Combined Forced and Scheduled Outage Analysis

7.3.1 Composite Outage Model

Realistic reliability assessment requires modeling both forced and scheduled outages simultaneously.

Unit Availability with Maintenance:

$$A_i(t) = \begin{cases} 0 & \text{if unit } i \text{ in scheduled maintenance during } t \\ 1 - \text{FOR}_i & \text{otherwise} \end{cases} \quad (105)$$

Time-Dependent COPT:

For each time period t :

1. Identify units available (not in scheduled maintenance)
2. Build COPT using only available units and their FOR values
3. Calculate reliability indices for period t
4. Aggregate across all time periods

Annual LOLP with Maintenance:

$$\text{LOLP}_{\text{annual}} = \frac{1}{T} \sum_{t=1}^T \text{LOLP}_t \quad (106)$$

where LOLP_t is calculated using capacity available in period t after maintenance scheduling.

7.3.2 Maintenance Risk Assessment**Maintenance Outage Contribution:**

Quantify reliability impact of maintenance scheduling:

$$\Delta \text{LOLP}_{\text{maintenance}} = \text{LOLP}_{\text{with maintenance}} - \text{LOLP}_{\text{forced only}} \quad (107)$$

$$\text{Maintenance Contribution} = \frac{\Delta \text{LOLP}_{\text{maintenance}}}{\text{LOLP}_{\text{with maintenance}}} \times 100\% \quad (108)$$

Optimal Maintenance Window Identification:

For each unit, evaluate reliability impact across potential maintenance windows:

$$\text{Risk Score}(i, t) = \sum_{s=t}^{t+d_i} \frac{L_s}{\text{Available Capacity}_s} \quad (109)$$

Select maintenance window that minimizes cumulative risk score while satisfying constraints.

8 Lecture 3.7: Interconnected System Analysis

Interconnected operation between multiple power systems provides significant reliability benefits through capacity and energy sharing. This lecture examines methods for evaluating interconnection benefits and modeling multi-area system reliability.

8.1 Interconnection Benefits

8.1.1 Types of Interconnection Benefits

Interconnection Reliability Benefits:

1. Capacity Sharing:

- Emergency assistance during capacity shortfalls
- Reduced individual system reserve requirements
- Access to diverse generation resources

2. Load Diversity:

- Peak demands occur at different times
- Reduces aggregate peak relative to sum of individual peaks
- Enables capacity optimization across areas

3. Generation Diversity:

- Uncorrelated forced outages reduce simultaneous failures
- Different maintenance schedules increase available capacity
- Complementary renewable resources (wind/solar patterns)

4. Economic Benefits:

- Access to lower-cost generation
- Economies of scale in large unit construction
- Reduced total installed capacity requirements

Diversity Interchange Capacity (DIC):

The reduction in required capacity due to load diversity:

$$\text{DIC} = \sum_{i=1}^N L_{i,\text{peak}} - L_{\text{system peak}} \quad (110)$$

where:

- $L_{i,\text{peak}}$ = Peak load of area i
- $L_{\text{system peak}}$ = Coincident system peak load

Example 8.1 (Load Diversity Benefit):

Consider three interconnected areas:

Area	Individual Peak (MW)	Time of Peak	Load at System Peak
Area A	1,000	2 PM	950
Area B	800	6 PM	700
Area C	1,200	8 PM	1,000
Total	3,000	-	2,650

Diversity Benefit Calculation:

$$\text{Sum of Individual Peaks} = 1,000 + 800 + 1,200 = 3,000 \text{ MW} \quad (111)$$

$$\text{Coincident System Peak} = 2,650 \text{ MW} \quad (112)$$

$$\text{Diversity Interchange Capacity} = 3,000 - 2,650 = 350 \text{ MW} \quad (113)$$

$$\text{Diversity Factor} = \frac{2,650}{3,000} = 0.883 = 88.3\% \quad (114)$$

Capacity Savings: If each area plans with 15% reserve margin in isolation:

$$\text{Total Isolated Capacity} = 3,000 \times 1.15 = 3,450 \text{ MW} \quad (115)$$

$$\text{Interconnected Capacity} = 2,650 \times 1.15 = 3,048 \text{ MW} \quad (116)$$

$$\text{Capacity Reduction} = 3,450 - 3,048 = 402 \text{ MW} \quad (117)$$

$$\text{Cost Savings} = 402 \times \$1,200/\text{kW} = \$482 \text{ million} \quad (118)$$

8.2 Interconnection Modeling**8.2.1 Tie Line Capacity Representation**

Interconnections are characterized by transfer capability limits that constrain assistance between areas.

Tie Line States:

Tie lines can fail, reducing interconnection capacity:

$$P_{\text{tie}}(t) = \begin{cases} P_{\text{tie}}^{\max} & \text{with probability } (1 - \text{FOR}_{\text{tie}}) \\ 0 & \text{with probability } \text{FOR}_{\text{tie}} \end{cases} \quad (119)$$

Typical tie line FOR values: 1-3% for transmission circuits.

8.2.2 Two-Area Reliability Model**Simplified Two-Area Formulation:**

For areas A and B with tie line capacity T_{AB} :

Area A adequacy condition:

$$C_A + \min(T_{AB}, \text{Surplus}_B) \geq L_A \quad (120)$$

where $\text{Surplus}_B = \max(0, C_B - L_B)$ is available assistance from Area B.

Composite System LOLP:

Must consider all combinations of:

- Area A capacity states
- Area B capacity states
- Tie line availability states
- Load levels in both areas

Probability of Deficiency in Area A:

$$P(\text{Deficiency}_A) = \sum_{C_A, C_B, T} P(C_A, C_B, T) \times \mathbb{I}[C_A + \min(T, C_B - L_B) < L_A] \quad (121)$$

Example 8.2 (Two-Area Interconnection Analysis):**System Data:**

Parameter	Area A	Area B
Installed Capacity	1,200 MW	1,000 MW
Peak Load	1,000 MW	800 MW
Average FOR	5%	5%

Tie line capacity: 200 MW, Tie line FOR: 2%

Isolated System Reliability:

Area A:

$$\text{Reserve Margin}_A = \frac{1,200 - 1,000}{1,000} = 20\% \quad (122)$$

$$\text{Expected Capacity}_A = 1,200 \times 0.95 = 1,140 \text{ MW} \quad (123)$$

$$\text{LOLP}_A^{\text{isolated}} \approx 0.002 \text{ (from COPT analysis)} \quad (124)$$

Interconnected System Reliability:

With tie line, Area A can receive up to 200 MW assistance from Area B when:

- Area B has surplus capacity: $C_B > 800 \text{ MW}$
- Tie line is available (98% probability)

Expected assistance to Area A:

$$E[\text{Assistance}] = 0.98 \times P(C_B > 800) \times E[\min(200, C_B - 800) | C_B > 800] \quad (125)$$

$$\approx 0.98 \times 0.95 \times 150 = 139.6 \text{ MW} \quad (126)$$

Improved reliability:

$$\text{LOLP}_A^{\text{interconnected}} \approx 0.0005 \text{ (75\% reduction)} \quad (127)$$

The interconnection significantly improves Area A reliability through access to Area B capacity.

8.3 Assistance Probability and Availability

8.3.1 Assistance Probability Function

The probability that assistance is available from neighboring areas depends on their capacity surplus conditions.

General Assistance Function:

$$P_{\text{assist}}(A_{\text{req}}) = \sum_{C_{\text{neighbor}}} P(C_{\text{neighbor}}) \times \mathbb{I}[C_{\text{neighbor}} - L_{\text{neighbor}} \geq A_{\text{req}}] \quad (128)$$

where:

- A_{req} = Requested assistance amount
- C_{neighbor} = Available capacity in neighboring area
- L_{neighbor} = Load in neighboring area

Assistance Probability Characteristics:

- Decreases as requested assistance increases
- Higher when neighboring area has high reserve margin
- Reduced by tie line limitations and failures
- Depends on correlation of load patterns between areas

8.3.2 Multi-Area System Analysis

For systems with N interconnected areas, comprehensive analysis requires:

1. Network Topology:

$$\text{Transfer Limit}_{ij} = f(\text{Network Configuration, Power Flows, Contingencies}) \quad (129)$$

2. Simultaneous Deficiency Probability:

$$P(\text{Multiple Areas Deficient}) = P\left(\bigcap_{i \in S} \{\text{Area } i \text{ Deficient}\}\right) \quad (130)$$

This probability is typically much lower than product of individual probabilities due to load diversity and generation diversity effects.

3. Optimal Assistance Allocation: When multiple areas experience deficiency, available assistance must be allocated:

$$\max \sum_i V_i \times A_i \quad (131)$$

$$\text{s.t.} \quad \sum_i A_i \leq A_{\text{available}} \quad (132)$$

$$A_i \leq D_i \quad \forall i \quad (133)$$

where:

- V_i = Value of lost load in area i
- A_i = Assistance allocated to area i
- D_i = Deficiency in area i

9 Lecture 3.8: Emergency Assistance and Contingency Planning

Emergency assistance protocols and contingency planning procedures ensure coordinated response to generation adequacy emergencies in interconnected systems. This lecture covers emergency operating procedures, assistance protocols, and contingency reserve requirements.

9.1 Emergency Operating Procedures

9.1.1 Operating Reserve Categories

Operating Reserve Classifications:

1. Regulating Reserve (Seconds Response):

- Automatic generation control (AGC) response
- Frequency regulation service
- Typical requirement: 1% of hourly load

2. Spinning Reserve (10-Minute Response):

- Synchronized generators with available headroom
- Immediate response capability
- Typical requirement: Largest single contingency

3. Non-Spinning Reserve (30-Minute Response):

- Quick-start generation (combustion turbines)
- Interruptible loads
- Typical requirement: Second largest contingency

4. Replacement Reserve (60+ Minute Response):

- Additional generation or demand response
- Restores spinning/non-spinning reserves
- Variable requirements based on system conditions

Total Operating Reserve Requirement:

$$R_{\text{total}} = R_{\text{reg}} + R_{\text{spin}} + R_{\text{non-spin}} + R_{\text{replacement}} \quad (134)$$

Typical values:

$$R_{\text{reg}} = 1\% \times L_{\text{forecast}} \quad (135)$$

$$R_{\text{spin}} = \max(\text{Largest Unit}, 3\% \times L_{\text{forecast}}) \quad (136)$$

$$R_{\text{non-spin}} = 50\% \times R_{\text{spin}} \quad (137)$$

$$R_{\text{replacement}} = R_{\text{spin}} + R_{\text{non-spin}} \quad (138)$$

9.2 Emergency Assistance Protocols

9.2.1 Coordinated Emergency Response

Emergency assistance between interconnected systems follows established protocols ensuring rapid and reliable response.

Emergency Priority Levels:

Level 1 - Energy Alert:

- Reserves below normal operating levels

- No load shedding anticipated
- Increased monitoring and coordination
- Pre-position emergency resources

Level 2 - Energy Emergency:

- Reserves critically low ($<$ minimum requirement)
- Load shedding possible within hours
- Request emergency assistance from neighbors
- Implement demand response programs
- Public appeals for conservation

Level 3 - Load Shedding in Progress:

- Controlled load shedding underway
- Maximum emergency assistance mobilized
- Curtailment of interruptible contracts
- Emergency operations center activated

9.2.2 Assistance Request and Response Protocol

Request Procedure:

1. System in emergency declares emergency level
2. Quantifies assistance requirement:

$$A_{\text{required}} = L_{\text{forecast}} + R_{\text{min}} - C_{\text{available}} \quad (139)$$

3. Broadcasts assistance request to interconnected neighbors
4. Specifies duration and urgency of need

Response Criteria:

Neighboring systems assess their ability to provide assistance:

$$A_{\text{available}} = \max(0, C_{\text{surplus}} - R_{\text{own}} - \text{Tie Limit}) \quad (140)$$

where:

- C_{surplus} = Available capacity minus own load
- R_{own} = Own reserve requirements (must maintain)
- Tie Limit = Transfer capability to requesting system

Assistance Provision Decision:

$$\text{Provide Assistance} = \begin{cases} \text{Yes} & \text{if } A_{\text{available}} > 0 \text{ and own reliability adequate} \\ \text{No} & \text{otherwise} \end{cases} \quad (141)$$

9.3 Load Shedding Procedures

9.3.1 Controlled Load Shedding Strategy

When emergency assistance is insufficient, controlled load shedding (rolling blackouts) prevents uncontrolled system collapse.

Load Shedding Principles:

1. **Staged Implementation:** Shed load in discrete blocks (5-10% of total)
2. **Equitable Distribution:** Rotate outages among customer areas
3. **Critical Load Protection:** Maintain service to hospitals, emergency services, critical infrastructure
4. **Duration Limitation:** Limit individual outage duration (typically 1-2 hours)
5. **System Frequency Maintenance:** Shed sufficient load to stabilize frequency

Automatic Under-Frequency Load Shedding (UFLS):

Emergency load shedding triggered automatically by low system frequency:

Frequency Threshold (Hz)	Load Shed (%)	Cumulative Shed (%)
59.5	5	5
59.3	5	10
59.0	5	15
58.7	5	20
58.5	10	30

Required Load Shedding Calculation:

To restore frequency balance:

$$\Delta P_{\text{shed}} = \frac{\Delta f}{K_{\text{system}}} \quad (142)$$

where:

- Δf = Frequency deviation from nominal (Hz)
- K_{system} = System frequency response characteristic (%/0.1 Hz)
- Typical K_{system} values: 1.0-2.0 %/0.1 Hz

Example 9.1 (Emergency Load Shedding Calculation):

System Conditions:

- System frequency: 59.3 Hz (deviation = -0.7 Hz)
- System load: 5,000 MW
- System frequency response: 1.5%/0.1 Hz
- Available generation: 4,600 MW

Required Load Shedding:

$$\text{Generation Deficit} = 5,000 - 4,600 = 400 \text{ MW} \quad (143)$$

$$\text{Additional Margin for Stability} = 100 \text{ MW} \quad (144)$$

$$\text{Total Load Shed Required} = 400 + 100 = 500 \text{ MW} = 10\% \text{ of load} \quad (145)$$

Load Shedding Implementation:

- Stage 1 (59.5 Hz trigger): 500 MW (10%)
- Shed in 4 geographic blocks of 125 MW each
- Rotate blocks every 60 minutes if emergency persists
- Protect critical facilities (hospitals, emergency services)

Expected Frequency Recovery:

$$f_{\text{new}} = 60 + \left(\frac{4,600 - (5,000 - 500)}{5,000} \right) \times 5 \times 0.1 = 59.8 \text{ Hz} \quad (146)$$

Additional generation or further load shedding may be required to fully restore nominal frequency.

9.4 Contingency Planning and Reserve Adequacy

9.4.1 Contingency Reserve Requirements

Systems must maintain sufficient reserves to respond to credible contingencies without load shedding.

N-1 Contingency Criterion:

System must withstand loss of largest single element:

$$R_{\text{requirement}} \geq \max_i \{C_i\} \quad (147)$$

where C_i is the capacity of generator unit i .

Probabilistic Reserve Requirement:

More sophisticated approach based on multiple contingency probabilities:

$$R_{\text{requirement}} = \sum_i C_i \times P(\text{Unit } i \text{ fails}) + \text{Load Forecast Error} \quad (148)$$

Typical formulation includes:

$$R_{\text{total}} = \sqrt{(k \times \sigma_L)^2 + \sum_{i=1}^N (C_i \times \sqrt{P_i(1 - P_i)})^2} \quad (149)$$

$$(150)$$

where:

- σ_L = Standard deviation of load forecast error
- k = Confidence level factor (typically 1.96 for 95%)
- P_i = Forced outage rate of unit i

9.4.2 Reliability-Based Reserve Optimization

Optimize reserve levels to balance reliability costs against reserve provision costs:

Objective Function:

$$\min \quad C_{\text{reserves}}(R) + C_{\text{reliability risk}}(R) \quad (151)$$

where:

$$C_{\text{reserves}}(R) = c_r \times R \quad (\text{linear reserve cost}) \quad (152)$$

$$C_{\text{reliability risk}}(R) = \text{VOLL} \times \text{EUE}(R) \quad (\text{expected outage cost}) \quad (153)$$

Optimal Reserve Level:

Found where marginal reserve cost equals marginal reliability benefit:

$$\frac{dC_{\text{reserves}}}{dR} = -\frac{dC_{\text{reliability risk}}}{dR} \quad (154)$$

💡 Practical Reserve Margin Guidelines:

Industry practice typically uses fixed reserve margin criteria:

- Planning reserve margin: 12-18% of peak load
- Operating reserve: 6-10% of current load
- Spinning reserve: Largest unit or 3% of load

These values are periodically validated through comprehensive reliability studies considering:

- Generation fleet characteristics (unit sizes, FOR)
- Load characteristics (variability, forecast uncertainty)
- Interconnection benefits and limitations
- Historical reliability performance
- Value of lost load and customer preferences

10 🎯 Module 3 Summary and Key Takeaways

10.1 Fundamental Concepts Mastered

Through comprehensive coverage of generation planning and reliability, students have developed deep understanding of:

1. **Generation Planning Framework:** Systematic approaches to capacity expansion considering economic, reliability, and environmental objectives
2. **Integrated Resource Planning:** Comprehensive evaluation of supply-side and demand-side resources using consistent methodologies
3. **Probabilistic Modeling:** Stochastic representation of generation system behavior using COPT and recursive algorithms
4. **Reliability Metrics:** LOLP, LOLE, LOLH, and EUE as quantitative measures of generation adequacy
5. **Capacity Expansion Optimization:** Multi-period decision-making frameworks balancing timing, technology, and risk considerations
6. **Maintenance Scheduling:** Optimal timing of planned outages to maintain reliability while minimizing costs

7. **Interconnection Benefits:** Quantification of reliability improvements from capacity sharing and load diversity
8. **Emergency Procedures:** Coordinated response protocols for generation adequacy emergencies

10.2 Practical Skills Developed

Analytical Capabilities:

- Construct and analyze capacity outage probability tables (COPT)
- Calculate generation system reliability indices (LOLP, EUE)
- Evaluate capacity expansion alternatives using economic and reliability criteria
- Optimize maintenance schedules considering system constraints
- Assess interconnection benefits quantitatively
- Design emergency response procedures and reserve requirements

Computational Proficiency:

- Implement recursive algorithms for large system analysis
- Develop load models for reliability studies
- Program Monte Carlo simulation methods
- Apply optimization techniques to capacity expansion problems
- Create decision support tools for planning studies

10.3 Industry Applications

The methodologies covered in this module directly support professional practice in:

- **Utility Planning Departments:** Resource adequacy assessments and integrated resource plans
- **Independent System Operators:** Reserve requirement determination and emergency procedures
- **Regulatory Agencies:** Evaluation of utility planning submissions and reliability standards
- **Consulting Firms:** Generation planning studies and reliability assessments for clients
- **Equipment Manufacturers:** Understanding customer requirements for reliability performance
- **Financial Institutions:** Risk assessment for generation project financing

10.4 Key Formulas and Relationships

Reliability Fundamentals:

$$\text{Availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} \quad (155)$$

$$\text{FOR} = \frac{\text{MTTR}}{\text{MTTF} + \text{MTTR}} \quad (156)$$

$$\text{LOLP} = \sum_{C < L} P(C) \quad (157)$$

$$\text{EUE} = \sum_{C < L} P(C) \times (L - C) \quad (158)$$

Economic Analysis:

$$\text{LCOE} = \frac{\sum_t \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_t \frac{E_t}{(1+r)^t}} \quad (159)$$

$$\text{NPV} = \sum_{t=0}^T \frac{\text{Cash Flow}_t}{(1+r)^t} \quad (160)$$

Reserve Requirements:

$$\text{Reserve Margin} = \frac{C_{\text{installed}} - L_{\text{peak}}}{L_{\text{peak}}} \quad (161)$$

$$R_{\text{spinning}} = \max(\text{Largest Unit}, 3\% \times L) \quad (162)$$

Interconnection Benefits:

$$\text{Diversity Factor} = \frac{L_{\text{coincident peak}}}{\sum_i L_{i,\text{peak}}} \quad (163)$$

$$\text{Capacity Savings} = \sum_i L_{i,\text{peak}} - L_{\text{coincident peak}} \quad (164)$$

10.5 Critical Learning Points

! Essential Concepts for Professional Practice:

1. **Reliability is Probabilistic:** Generation adequacy must be evaluated using probabilistic methods; deterministic criteria provide limited insight into actual system risk.
2. **Planning Requires Multiple Objectives:** Economic optimization alone is insufficient; reliability, environmental, and social objectives must be balanced systematically.
3. **Uncertainty is Fundamental:** Load forecasts, equipment failures, fuel prices, and technology performance all involve uncertainty requiring robust decision-making frameworks.
4. **Interconnection Provides Value:** Coordinated planning and operation of interconnected systems significantly reduces capacity requirements and improves reliability.
5. **Maintenance Scheduling Matters:** Optimal timing of planned outages can substantially impact system reliability without additional capital investment.
6. **Reserves Are Essential:** Adequate operating reserves enable system response to contingencies and prevent cascading failures.
7. **Demand-Side Resources Count:** Energy efficiency and demand response can provide capacity and reliability benefits at lower cost than supply-side resources.
8. **Technology Mix Affects Risk:** Portfolio diversification across technologies and fuels reduces exposure to price volatility and supply disruptions.

10.6 Further Study and Advanced Topics

Students interested in deeper exploration of generation planning should investigate:

- **Advanced Reliability Methods:** Monte Carlo simulation, frequency and duration analysis, composite system reliability
- **Renewable Integration:** Variable generation modeling, capacity credit assessment, flexibility requirements
- **Market Mechanisms:** Capacity markets, resource adequacy requirements in restructured systems
- **Risk Analysis:** Real options analysis, decision analysis under uncertainty, robust optimization
- **Climate Change Adaptation:** Long-term planning under climate uncertainty, extreme weather impacts
- **Energy Storage:** Optimal storage sizing and operation, storage contribution to reliability
- **Distributed Resources:** Aggregated DER contributions to system adequacy, hosting capacity analysis

11 Programming Assignment 3: Generation Planning Optimization

11.1 Assignment Overview

This assignment requires development of comprehensive algorithms for generation system reliability assessment and capacity planning optimization. Students will implement LOLP calculation methods, capacity expansion optimization, and maintenance scheduling algorithms.

11.2 Assignment Components

11.2.1 Part 1: COPT Generation and LOLP Calculation

Objectives:

- Implement recursive algorithm for COPT construction
- Calculate LOLP, LOLE, and EUE for generation systems
- Analyze sensitivity to unit FOR and capacity levels
- Validate results against analytical solutions

Required Functions:

```

1  def build_copt_recursive(units):
2      """
3      Build Capacity Outage Probability Table
4
5      Parameters:
6      -----
7      units : list of dict
8      Each unit: {'capacity': MW, 'FOR': fraction}
9
10     Returns:
11     -----
12     copt : dict
13     {capacity: probability}
14     """
15     pass
16
17     def calculate_lolp(copt, peak_load):
18         """Calculate Loss of Load Probability"""
19         pass
20
21     def calculate_eue(copt, load_levels, durations):
22         """Calculate Expected Unserved Energy"""
23         pass
24

```

Listing 1: Required LOLP Functions

11.2.2 Part 2: Capacity Expansion Optimization

Objectives:

- Formulate capacity expansion as optimization problem
- Implement solution algorithm (linear programming or heuristic)
- Evaluate multiple technology options

- Perform sensitivity analysis on key parameters

Problem Formulation:

- Decision variables: Capacity additions by technology and year
- Objective: Minimize net present value of costs
- Constraints: Reliability criteria, energy balance, capacity limits
- Planning horizon: 10-20 years

11.2.3 Part 3: Maintenance Scheduling Optimization

Objectives:

- Develop maintenance scheduling algorithm
- Minimize reliability impact while satisfying maintenance requirements
- Handle crew and resource constraints
- Produce practical implementation schedules

11.3 Deliverables

1. **Source Code:** Well-documented Python or MATLAB implementation
2. **Test Cases:** Comprehensive test suite validating algorithm correctness
3. **Case Study Analysis:**
 - 10-unit generation system reliability assessment
 - 20-year capacity expansion plan
 - Annual maintenance schedule optimization
4. **Technical Report:** 15-20 pages covering:
 - Algorithm description and justification
 - Implementation details and data structures
 - Results presentation and interpretation
 - Sensitivity analysis findings
 - Conclusions and recommendations
5. **User Documentation:** Instructions for using developed tools

11.4 Evaluation Criteria

Criterion	Weight	Description
Algorithm Correctness	30%	Accurate implementation of methods
Code Quality	15%	Documentation, structure, efficiency
Case Study Analysis	25%	Depth and quality of results
Technical Report	20%	Clarity, completeness, professionalism
Creativity	10%	Novel approaches or extensions
Total	100%	

11.5 Submission Requirements

Due Date: Three weeks from assignment distribution

Submission Format:

- Single ZIP file containing all materials
- Code in separate folder with README
- Report as PDF document
- Test data and results in organized structure

File Naming Convention: PSM104_Assignment3_RollNumber_Name.zip

12 References and Further Reading

12.1 Primary References

1. R. Billinton and R. N. Allan, *Reliability Evaluation of Power Systems*, 2nd ed. New York: Plenum Press, 1996.
2. A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. New York: Wiley, 2013.
3. R. L. Sullivan, *Power System Planning*, New York: McGraw-Hill, 1977.
4. IEEE Task Force on Capacity Benefits, "Capacity credit of generation," *IEEE Trans. Power Syst.*, 2021.

12.2 Advanced Reading

1. M. Shahidehpour, H. Yamin, and Z. Li, *Market Operations in Electric Power Systems*, Wiley-IEEE Press, 2002.
2. R. Billinton and W. Li, *Reliability Assessment of Electric Power Systems Using Monte Carlo Methods*, Plenum Press, 1994.
3. G. Strbac et al., "Impact of wind generation on the operation and development of the UK electricity systems," *Electric Power Systems Research*, 2007.

12.3 Industry Standards

1. NERC, "Probabilistic Assessment Technical Guideline Document," North American Electric Reliability Corporation, 2018.
2. IEEE Std 762-2006, "IEEE Standard Definitions for Use in Reporting Electric Generating Unit Reliability, Availability, and Productivity."
3. IEEE PES Resource Adequacy Task Force, "Probabilistic Methods for Planning Reserve Assessment," 2020.
4. CIGRÉ Technical Brochure, "Probabilistic Methods for Power System Reliability Evaluation," Working Group C4.601, 2015.

13 Practice Problems for Module 3

13.1 Problem Set 1: Basic Reliability Calculations

Problem 13.1 (Two-State Unit Reliability):

A 250 MW coal-fired generating unit has the following reliability data:

- Mean Time To Failure (MTTF) = 1,500 hours
- Mean Time To Repair (MTTR) = 80 hours

Calculate:

1. Availability and Forced Outage Rate
2. Expected available capacity over a one-year period
3. Probability the unit is operational at any given time

Problem 13.2 (Three-Unit System LOLP):

A generation system consists of three identical units with the following characteristics:

- Unit capacity: 150 MW each
- Availability: 0.96
- System peak load: 400 MW

Determine:

1. Complete Capacity Outage Probability Table (COPT)
2. Loss of Load Probability (LOLP)
3. Loss of Load Expectation (LOLE) in days/year
4. Expected Unserved Energy (EUE)

13.2 Problem Set 2: Capacity Expansion Planning

Problem 13.3 (LCOE Comparison):

Compare two generation technologies for baseload operation:

Option A - Coal Plant:

- Capital cost: \$3,500/kW
- Fixed O&M: \$45/kW-year
- Variable O&M: \$4.5/MWh
- Heat rate: 9,000 Btu/kWh
- Coal price: \$2.50/MMBtu
- Capacity factor: 85%

- Economic life: 40 years

Option B - Nuclear Plant:

- Capital cost: \$6,500/kW
- Fixed O&M: \$100/kW-year
- Variable O&M: \$2.0/MWh
- Heat rate: 10,400 Btu/kWh
- Uranium cost: \$0.70/MMBtu
- Capacity factor: 92%
- Economic life: 60 years

Using a discount rate of 7%, calculate:

1. Levelized Cost of Energy (LCOE) for each option
2. Break-even coal price where options are economically equivalent
3. Break-even capacity factor for nuclear plant

Problem 13.4 (Reserve Margin Analysis):

A power system has the following characteristics:

- Peak load: 3,500 MW
- Installed capacity: 4,200 MW
- Average unit FOR: 6%
- Largest single unit: 500 MW
- Target LOLE: 0.1 days/year

Analyze:

1. Current planning reserve margin
2. Expected available capacity during peak
3. Whether system meets reliability criterion
4. Additional capacity required to meet criterion (if needed)

13.3 Problem Set 3: Maintenance Scheduling

Problem 13.5 (Optimal Maintenance Schedule):

Schedule annual maintenance for a four-unit system:

Unit	Capacity (MW)	FOR	Maintenance Duration
1	300	0.05	4 weeks
2	300	0.05	4 weeks
3	250	0.04	3 weeks
4	250	0.04	3 weeks

Seasonal Load Profile:

- Winter (Weeks 1-13, 48-52): 950 MW peak
- Spring (Weeks 14-26): 700 MW peak
- Summer (Weeks 27-39): 1,000 MW peak
- Fall (Weeks 40-47): 750 MW peak

Required reserve margin: 15%

Develop an optimal maintenance schedule that:

1. Maintains adequate reserve margin in all periods
2. Minimizes peak period maintenance
3. Provides equitable distribution across seasons

13.4 Problem Set 4: Interconnection Analysis

Problem 13.6 (Two-Area Interconnection Benefits):

Two power systems are considering interconnection:

Area A:

- Installed capacity: 1,500 MW
- Peak load: 1,200 MW (occurs 2 PM)
- Average FOR: 5%
- Isolated LOLP: 0.005

Area B:

- Installed capacity: 1,200 MW
- Peak load: 1,000 MW (occurs 8 PM)
- Average FOR: 5%
- Isolated LOLP: 0.008

Proposed Interconnection:

- Tie line capacity: 200 MW

- Tie line FOR: 2%
- Load diversity factor: 0.90

Calculate:

1. Diversity interchange capacity
2. Potential capacity savings for each area
3. Expected improvement in reliability (qualitative)
4. Economic justification for tie line (assume \$1,000/kW capital cost)

Appendix A: Probability and Statistics Review

Essential Probability Concepts

Basic Probability Axioms:

$$0 \leq P(A) \leq 1 \quad \text{for any event } A \quad (165)$$

$$P(\Omega) = 1 \quad (\text{sample space}) \quad (166)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (167)$$

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (168)$$

Independent Events:

$$P(A \cap B) = P(A) \times P(B) \quad \text{if } A \text{ and } B \text{ independent} \quad (169)$$

Binomial Distribution:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (170)$$

Expected Value and Variance:

$$E[X] = \sum_i x_i P(x_i) \quad (171)$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad (172)$$

Common Distributions in Reliability

Exponential Distribution (failure times):

$$f(t) = \lambda e^{-\lambda t} \quad (173)$$

$$F(t) = 1 - e^{-\lambda t} \quad (174)$$

$$R(t) = e^{-\lambda t} \quad (175)$$

$$E[T] = \frac{1}{\lambda} = \text{MTTF} \quad (176)$$

Normal Distribution (load forecasting):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (177)$$

Appendix B: Nomenclature and Abbreviations

Acronyms

CAIDI	Customer Average Interruption Duration Index
CCGT	Combined Cycle Gas Turbine
COPT	Capacity Outage Probability Table
DIC	Diversity Interchange Capacity
ELCC	Effective Load Carrying Capability
EENS	Expected Energy Not Served
EUE	Expected Unserved Energy
FOR	Forced Outage Rate
HHI	Herfindahl-Hirschman Index
IRP	Integrated Resource Planning
LCOE	Levelized Cost of Energy
LOLE	Loss of Load Expectation
LOLH	Loss of Load Hours
LOLP	Loss of Load Probability
MTTF	Mean Time To Failure
MTTR	Mean Time To Repair
NPV	Net Present Value
OCGT	Open Cycle Gas Turbine
SAIDI	System Average Interruption Duration Index
SAIFI	System Average Interruption Frequency Index
UFLS	Under-Frequency Load Shedding
VOLL	Value of Lost Load

Symbols

A	Availability (fraction)
C	Capacity (MW)
CF	Capacity Factor (fraction)
D	Demand or Load (MW)
E	Energy (MWh)
f	Frequency (Hz) or probability density function
F	Cumulative distribution function
IC	Investment Cost (\$/kW)
L	Load (MW)
P	Probability or Power (MW)
r	Discount rate (fraction)
R	Reserve (MW) or Reliability function
t	Time (hours, years)
T	Planning horizon (years) or Tie line capacity (MW)
U	Unavailability or Forced Outage Rate
λ	Failure rate (failures/hour)
μ	Repair rate (repairs/hour)
σ	Standard deviation

Appendix C: Software Tools and Resources

Recommended Software for Generation Planning

Open Source Tools:

- **Python Libraries:**
 - NumPy/SciPy: Numerical computations
 - Pandas: Data analysis and manipulation
 - Matplotlib/Seaborn: Visualization
 - PuLP/Pyomo: Optimization modeling
- **MATLAB:**
 - Optimization Toolbox
 - Statistics and Machine Learning Toolbox
 - Simulink for dynamic system modeling
- **R Statistical Software:**
 - Reliability analysis packages
 - Time series forecasting tools
 - Advanced statistical methods

Commercial Software:

- PLEXOS: Integrated energy modeling
- PROMOD: Production cost modeling
- PSS/E: Power system simulation
- PowerWorld: Power flow and reliability analysis

Online Resources

- **IEEE Power & Energy Society:** Technical publications and standards
- **NERC:** Reliability standards and assessment reports
- **EIA:** Energy data and forecasting resources
- **IEA:** International energy statistics and analysis
- **EPRI:** Research reports on power system planning

Datasets for Practice

- Historical load data: ISO/RTO websites
- Generation unit characteristics: EIA Form 860
- Fuel prices: EIA energy price databases
- Renewable resources: NREL databases and tools
- Weather data: NOAA archives

Concluding Remarks

Generation planning and reliability assessment represent critical functions in electric power system development and operation. The methodologies covered in this module provide the analytical foundation necessary for ensuring adequate, reliable, and economical electricity supply.

Key Principles for Professional Practice:

1. **Systematic Analysis:** Apply structured analytical frameworks consistently
2. **Probabilistic Thinking:** Embrace uncertainty and quantify risks appropriately
3. **Multi-Objective Balance:** Consider reliability, economics, and environmental factors holistically
4. **Stakeholder Engagement:** Incorporate diverse perspectives in planning decisions
5. **Continuous Learning:** Stay current with evolving technologies and methodologies
6. **Professional Ethics:** Prioritize public interest and system reliability

The electric power industry continues to evolve rapidly with renewable energy integration, market restructuring, digitalization, and climate change adaptation. The fundamental principles and analytical methods covered in this module provide a solid foundation for addressing these contemporary challenges while maintaining the reliability essential to modern society.

Students completing this module possess essential competencies for careers in utility planning, system operations, regulatory analysis, consulting, and energy policy development. Continued professional development through practical experience, advanced study, and engagement with industry developments will further enhance these capabilities.

End of Module 3 Class Notes

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